

DIGVIJAY SINGH

Engineering Manager | Engineering Lead | Tech Lead

Gurugram, India | +91-8076489924 | rathore.digvijay10@gmail.com | [linkedin.com/in/digvijay-rathore](https://www.linkedin.com/in/digvijay-rathore) | rathore-digvijay.github.io

SUMMARY

Engineering leader with 9 years of experience building and scaling backend platforms in insurtech. Currently lead 8 engineers across two charters at InsuranceDekho — AI Labs and the Renewal Platform — owning architecture, technical roadmap, hiring, and performance for 4 microservices and 2 products that process lakhs of policy lifecycle events every month. Defined the company's applied-AI charter and took a production LLM agent from zero to 9,000+ partner users. Deep hands-on expertise in Node.js/TypeScript, NestJS, Python, distributed systems, event-driven architecture, and MongoDB at scale.

CORE SKILLS

Engineering Leadership: Team Leadership, Hiring & Interview Panels, Performance Reviews, Mentoring, Technical Roadmaps, Stakeholder Management, Agile Delivery

Architecture: Distributed Systems, Microservices, Event-Driven Architecture, System Design (HLD/LLD), API Design, Scalability, High Availability, Fault Tolerance

Backend: Node.js, NestJS, Express.js, TypeScript, JavaScript, Python, FastAPI, REST APIs, WebSockets

AI / GenAI: LangGraph, RAG Pipelines, pgvector, Hybrid Search (BM25 + Vector), LLM Function Calling, Prompt Engineering, OpenAI & Claude APIs, Langfuse, Evals

Databases: MongoDB (schema design, indexing, sharding, replication), PostgreSQL, MySQL, Redis, Elasticsearch

Cloud & Infrastructure: AWS (EC2, S3, SQS, RDS, EKS), Docker, Kubernetes, RabbitMQ, Jenkins, CI/CD

Observability: Grafana, Prometheus, New Relic, Kibana, Graylog, Langfuse

PROFESSIONAL EXPERIENCE

InsuranceDekho (Girnarsoft), Gurugram

Apr 2022 – Present

Tech Lead — AI Labs & Renewal Platform (SDE-III → Tech Lead)

- Lead **8 engineers** across AI Labs and the Renewal Platform (up to 10 previously), owning hiring, performance reviews, promotions, mentoring, and delivery; multiple mentees promoted during tenure.
- Hired **5 engineers** and ran **45+ technical interview panels** including campus recruitment; set code review, design review, and engineering quality standards for the group.
- Own the technical roadmap for **4 microservices and 2 products**; partner with Product, Business, QA, and DevOps to translate BRDs/PRDs into scalable designs and push back on non-scalable decisions.
- Cut **AWS infrastructure cost ~10%** via HPA tuning, runtime upgrades, and cross-AZ transfer optimization; introduced rate limiting and distributed caching to protect insurer integrations during traffic spikes.

AI Labs — Conversational AI Platform for B2B Partners

Jan 2026 – Present

- Defined and own the **applied-AI charter** for the organization; built AI Labs from the ground up with a team of 4 and shipped the first production LLM agent in the company.
- Motor Quote Agent (live): end-to-end natural-language journey covering quote generation, plan comparison, commission/payout grid lookup, and policy issuance across **6+ insurers** (Kotak, ICICI, Reliance, Bajaj, HDFC, Acko).
- Adoption in a 15-day window: **26.5K conversations, 9.2K unique partner agents, 297K user messages**; ~\$0.02 average LLM cost per conversation, sub-second time-to-first-token, and 3–5s p90 response.
- Built the RAG knowledge base on **PostgreSQL + pgvector** with hybrid retrieval (BM25 + vector similarity, RRF fusion) for Make-Model-Variant resolution; HNSW/IVFFlat indexing and embedding ingestion pipelines.
- Designed stateful multi-step agent workflows in **LangGraph** with LLM function calling for entity extraction, tiered model routing (lightweight for intent, stronger for reasoning), and Langfuse-based tracing and evals.
- Stack: Python, FastAPI, LangGraph, PostgreSQL/pgvector, MongoDB, Redis, Langfuse, Kubernetes*

Renewal Platform — Motor, Health & SME

Dec 2023 – Present

- Own end-to-end architecture for renewals — lead creation, nudges, payments, and booking — processing **~3 lakh motor and ~50K health renewal leads per month** at **99.95% service uptime**.
- Drove a **~20% increase in renewal conversion** through journey redesign, targeted multi-channel nudges, and a single-click renewal flow that added ~5% lead conversion on its own.

- Architected a fault-tolerant **multi-channel communication engine** (WhatsApp, SMS, Email) with payment links and funnel tracking, on RabbitMQ event-driven workflows with retries, dead-letter queues, and idempotent processing.
- Built the **Health Renewals LMS used by 200+ calling agents** — automated lead assignment, disposition workflows, reporting, and a React.js ops dashboard giving real-time success/failure visibility and in-dashboard retry.
- Integrated the Arrowhead voicebot to automate **~500 outbound renewal calls/day**, reducing outreach cost ~30–40% and manual effort ~30%.
- Shipped a Python renewal-notice PDF parser that auto-tags policy documents at **~80% accuracy**; modelled MongoDB schemas with indexing, sharding, and replication for high-volume lifecycle data, with Redis caching on hot paths.

Health Shield & Centralized KYC

Nov 2022 – Jan 2024

- Led cross-functional delivery (backend + frontend) of **Health Shield**, an OPD/IPD non-insurance product opening a new revenue line for B2B agents — scaled to **~500 policies/day at peak** from 100–200/day at launch.
- Led a **centralized KYC platform** across 20 insurer integrations to meet IRDAI guidelines within a hard **45-day compliance deadline**; delivered as a reusable service consumed by every product vertical.

Wrocius Technologies (deputed to InsuranceDekho)

Jan 2021 – Apr 2022

Senior Software Engineer

- Led end-to-end development of the **Health Insurance marketplace**, onboarding 10+ insurers and 50+ plans into a unified platform through an adapter-based integration layer (Node.js, TypeScript, Express).
- Designed the rider/add-on journey for Health and Personal Accident products from scratch, increasing engagement on the quotes page.
- Drove **monolith-to-microservices migration** with Docker, Kubernetes, and CI/CD — cutting deployment time **~40%** with zero-downtime releases.
- Optimized queries and introduced Redis quote caching, improving retrieval performance **~20–30%**; built a role-based CMS for insurance content operations.

Creatiosoft Solutions, Noida

Sep 2017 – Jan 2021

Software Developer

- Built backend and game-server systems for a real-money multiplayer Poker platform (Node.js, WebSockets, Redis, MongoDB, MySQL) supporting **10K+ concurrent users**.
- Designed matchmaking, leaderboard, and pub/sub services plus load-testing bots; optimized queries and caching to improve response time **~35%**.
- Led a team of 4 on game-server architecture, delivery, and production support.

EDUCATION

IIMT College of Engineering, AKTU

2013 – 2017

Bachelor of Technology, Computer Science and Engineering